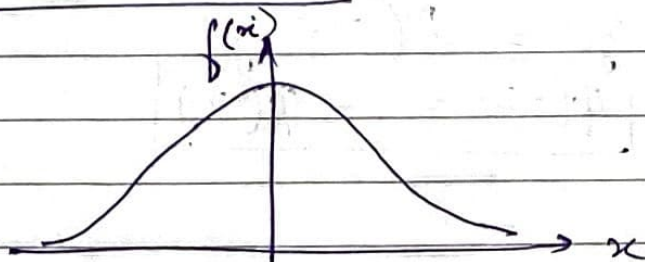


B.4 Normal Approximations

L1

Normal Distribution



for std. normal,

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

→ makes the
std. deviaⁿ = 1

A std. normal Z has $\mu=0, \sigma^2=1$

- What if I want Y to be normal with μ, σ^2 .

$$Z \sim N(0, 1)$$

$$\boxed{Y = \sigma Z + \mu} \sim N(\mu, \sigma^2)$$

$$E[Y] = \mu, \quad \text{Var}[Y] = \sigma^2$$

$$f_Y(y) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-\mu)^2}{2\sigma^2}}$$

- If, $Z \sim N(0, 1)$ $\boxed{\Phi_Z = P(Z \leq z)}$

CDF of Z

→ for this we
rely on tables

- Note - Tables for $\phi(z)$ only include $z \geq 0$. To find the CDF for negative values, we compare $\phi(-z)$ with $\phi(z)$.

$$P(Z > z) = P(Z \leq -z)$$

$$\therefore \boxed{\phi(-z) = 1 - \phi(z)}$$

eg. $X \sim N(2, 9)$

$$P(X > 3.86) = P\left(\frac{X - \mu}{\sigma} > \frac{3.86 - 2}{3}\right) = P(Z > 0.62)$$

$$1 - P(Z \leq 0.62) = 1 - 0.7324 = 0.2676$$

$$P(X > 1.49) = P\left(\frac{X - \mu}{\sigma} > \frac{1.49 - 2}{3}\right) = P(Z > -0.17) =$$

$$1 - \phi(-0.17) = \phi(0.17) = 0.5675$$

eg. $X \sim N(2, 9)$. Find 95th percentile of X .

$$0.95 = P(Z \leq z) = \phi(z)$$

$$z = \phi^{-1}(0.95)$$

$$z = 1.6449$$

→ the list of selected values increase at the bottom of Table.

$$X = \sigma z + \mu$$

$$t = 6.935$$

- If X & Y are independent Normals, it turns out $X+Y$ is also normal.

$$E(X+Y) = E(X) + E(Y)$$

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$$

$$X+Y \sim N(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2)$$

$$\frac{X+Y}{2} \sim N\left(\frac{\mu_X + \mu_Y}{2}, \frac{1}{4}(\text{Var}(X) + \text{Var}(Y))\right)$$

$$cX \sim N(cE(X), c^2 \text{Var}(X))$$

L2 Linear Interpolation

- $X \sim N(4, 2.2)$. $P(X > 3.3)$ to nearest 1000th.

$$P(X > 3.3) = P\left(\frac{X - \mu}{\sigma} > \frac{3.3 - 4}{\sqrt{2.2}}\right)$$

$$= 1 - P\left(Z \leq \frac{-0.7}{\sqrt{2.2}}\right)$$

$$= 1 - \Phi(-0.472) = \Phi(0.472)$$

$$\Phi(0.47) = 0.6808$$

$$\Phi(0.48) = 0.6844$$

$$\downarrow$$

0.682

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We can apply linear interpolation,

$$\frac{\Phi(t) - 0.6808}{t - 0.47} \approx \frac{0.6844 - 0.6804}{0.48 - 0.47}$$

$$\Phi(0.472) \approx 0.2(0.6844 - 0.6804) + 0.6808 \\ = 0.6815$$

L3 CLT (Central Limit Theorem)

- Idea $\rightarrow$ sum of large no. of RVs are often approximately normal.

If $X_1, \dots, X_n$ are iid RVs, then

$$\frac{(X_1 + \dots + X_n) - n E[X_1]}{\sqrt{n \text{Var}(X_1)}} \Rightarrow N(0, 1)$$

$$S_n = X_1 + X_2 + \dots + X_n$$

$$E[S_n] = n E[X_1] = n\mu$$

$$\text{Var}(S_n) = n \cdot \text{Var}(X_1) = n\sigma^2$$

$$\text{SD}(S_n) = \sigma\sqrt{n}$$

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} \Rightarrow N(0, 1)$$

$$\boxed{P(S_n \leq x) \approx \Phi\left(\frac{x - n\mu}{\sigma\sqrt{n}}\right)}$$

- $X_1, \dots, X_n$ are iid RVs. Let $\bar{X}$ be their avg.

$$S = X_1 + \dots + X_n$$

$$\bar{X} = \frac{S}{n} = \frac{1}{n} (X_1 + \dots + X_n)$$

$$E[\bar{X}] = E[X]$$

$$\text{Var}(\bar{X}) = \frac{\text{Var}(X)}{n}$$

- Since, S is roughly normal, so is S/n , so,

$$\bar{X} \approx N\left(E[X], \frac{\text{Var}(X)}{n}\right)$$

L4 Continuity Correcⁿ

- X is binomial $(25, 0.2)$

(a) $E[X] = 5$

$\text{Var}(X) = 4$

(b) $P(X \geq 8) = \sum_{k=8}^{25} P(X=k) = \sum_{k=8}^{25} {}^{25}C_k (0.2)^k (0.8)^{25-k}$

(c) $P(X > 8) = \sum_{k=9}^{25} P(X=k) = \sum_{k=9}^{25} {}^{25}C_k (0.2)^k (0.8)^{25-k}$

- A continuity correction is used with normal approximations of integer valued variables.

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It corrects for the normal being continuous while the variable we care about is not.

- It is used when approximating a sum of S_n discrete variables with a normal W .

$$P(S_n \in A) \approx P(W \text{ rounds to sth. } \in A)$$

1. $P(S_n < x) \approx P(W < x - 0.5)$
2. $P(S_n > x) \approx P(W > x + 0.5)$
3. $P(S_n \leq x) \approx P(W \leq x + 0.5)$
4. $P(S_n \geq x) \approx P(W \geq x - 0.5)$

L5 Lognormal R.V.

- Y is a lognormal if $Y = e^X$, X is normal.

↳ The log of lognormal distribution is normal.

$$\left. \begin{aligned} E[Y] &= E[e^X] = e^{\mu + \sigma^2/2} \\ E[Y^2] &= e^{2\mu + 2\sigma^2} \end{aligned} \right\} \begin{aligned} &X \sim N(\mu, \sigma^2) \\ &\text{with } Y = e^X \\ &\text{corresponding} \\ &\text{lognormal.} \end{aligned}$$

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- Ex. $Y \sim \text{Lognormal}(50, 400)$

$$P(Y > 75):$$

$$Y = e^X$$

$$E[Y] = E[e^X]$$

$$\Rightarrow 50 = e^{\mu + \sigma^2/2}$$

$$E[Y^2] = E[e^{2X}]$$

$$\Rightarrow \ln(50) = \frac{\mu + \sigma^2}{2}$$

$$\ln(50^2 + 400) = 2\mu + 2\sigma^2$$

$$\mu = 3.84, \quad \sigma^2 = 0.148$$

$$P(Y > 75) = P(X > \ln(75))$$

$$= P\left(\frac{X - \mu}{\sigma} > \frac{\ln(75) - 3.84}{\sqrt{0.148}}\right)$$

$$= 1 - \Phi(1.245) \approx 0.11$$